Bilingual Speech Synthesis with FastSpeech 2: An English and Telugu Study

Technical Report: English (Svarah) and Telugu (IndicTTS Telugu) Neural Text-to-Speech Systems

1. Overview

This project develops a non-autoregressive neural text-to-speech (TTS) workflow using FastSpeech 2 for two languages: English, using the Svarah corpus, and Telugu, using the IndicTTS Telugu corpus. It is based on the open-source PyTorch FastSpeech 2 implementation and is configured separately for each language through its dataset, preprocessing procedures, and output directories.

The model is designed to learn phoneme durations, fundamental frequency (F0), loudness patterns, and the mel-frequency spectrogram in one feed-forward operation. A HiFi-GAN neural vocoder then converts the spectrogram into an audio waveform. Project outputs include language-wise configuration files, generated speech, prosody-predictor visualizations, and five test utterances for each language evaluated for pronunciation and naturalness.

2. System Design and Method

2.1 Core Implementation

The solution is built on the commonly used xcmyz PyTorch version of FastSpeech 2. Its processing sequence contains:

- A transformer-based phoneme encoder that represents the incoming phoneme sequence;
- A prosody, or variance, predictor that estimates phoneme duration, F0, and loudness;
- A spectrogram decoder that generates the mel-frequency spectrogram, followed by a post-net stage for refinement;
- A HiFi-GAN vocoder that converts the spectrogram into the final waveform.

The implementation also preserves the practical choices specified in the reference configuration:

- F0 and loudness are estimated at phoneme level instead of frame level;
- F0 and loudness targets are normalized during data preparation;
- A six-layer decoder is used in place of the four-layer decoder described in the original paper, providing greater modelling capacity;
- Gradient clipping and learning-rate warm-up are applied to make optimization more stable;
- Speaker embeddings support training with multiple speakers.

2.2 Architecture Configuration

The configuration files retain the same reference architecture in both language directories. The principal settings are listed below.

Parameter	Value	Parameter	Value
transformer.encoder_layer	4	transformer.encoder_hidden	256
transformer.decoder_layer	6	transformer.decoder_hidden	256
transformer.encoder_head	2	transformer.decoder_head	2
variance_predictor.filter_size	256	variance_predictor.kernel_size	3
variance_embedding.pitch_quantization	linear	variance_embedding.energy_quantization	linear
vocoder.model	HiFi-GAN	multi_speaker	true

These settings are consistent with English/config/README.md and Telugu/config/README.md. Both documents emphasize phoneme-level prosody estimation and the use of a deeper decoder.

2.3 Adaptations for Each Language

The same neural network is applied to English and Telugu. Differences are limited to preprocessing and text-cleaning functions, dataset directories, validation-set size, language identifier, lexicon and text-normalization rules, and the locations used to save

generated audio and prosody plots.

Setting	English	Telugu
language	en	te
text_cleaners	english_cleaners	english_cleaners
val_size	691	525

Although the network structure is shared, Telugu presents a more demanding modelling task. Its writing system is morphologically rich and syllabic, causing greater variation in phoneme duration and prosody than English.

3. Data, Preprocessing, and Training Setup

3.1 English Dataset

The English configuration, English/config/English/preprocess.yaml, reads source files from `/kaggle/temp/work/raw_data/English` and saves prepared data to `/kaggle/temp/work/preprocessed_data/English`. Its main settings are language en, a 22,050 Hz sampling rate, 80 mel channels, normalized phoneme-level F0 and loudness, and 691 validation utterances held out from training. This arrangement is appropriate for a relatively clean and consistent dataset with a substantial validation portion for model selection. Phoneme alignment uses Montreal Forced Aligner (MFA) with the lexicon `lexicon/librispeech-lexicon.txt`.

3.2 Telugu Dataset

The Telugu configuration, Telugu/config/Telugu/preprocess.yaml, follows the same settings while using `/kaggle/temp/work/raw_data/Telugu` as its source and `/kaggle/temp/work/preprocessed_data/Telugu` for processed output. The language identifier is te, and the validation partition contains 525 utterances. The preparation procedure remains the same, although the data are expected to have greater acoustic and phonological complexity because of vowel-consonant combinations, spelling variation, and prosodic phrasing.

3.3 Data Preparation Procedure

Both languages use the standard FastSpeech 2 preparation workflow:

- 1. Execute prepare_align.py to prepare the dataset for alignment;
- 2. Generate phoneme-level alignments with MFA or use alignment files that have already been generated;
- 3. Execute preprocess.py to obtain mel spectrograms, F0, loudness, phoneme durations, and text/phoneme sequences;
- 4. Save the resulting files in the appropriate preprocessed_data/... directory.

Audio / feature setting	Value
Sampling rate	22,050 Hz
STFT filter length / window length	1024 / 1024
Hop length	256
Mel channels	80
F0 and loudness resolution	Phoneme level, normalized

The configuration notes favour phoneme-level feature estimation because it produces more natural prosody than frame-level prediction.

3.4 Training and Optimization Schedule

The train.yaml files for both languages use almost identical optimization settings:

Hyperparameter	Value	Hyperparameter	Value
batch_size	16	total_step	40000
grad_clip_thresh	1.0	save_step	10000
weight_decay	0.0	val_step	1000
warm_up_step	4000	anneal_steps	[28000, 34000]
anneal_rate	0.3		

Training begins with a warm-up period, followed by learning-rate reductions at later stages. Model checkpoints and log files are saved in separate directories for each language.

4. Loss-Curve Analysis

Loss visualizations are available in Results/English/ and Results/Telugu/. Each directory contains training_loss.png and validation_loss.png.

4.1 Convergence Pattern

- Loss decreases sharply during the first few thousand training steps;
- Improvement becomes slower but remains steady in the later stages;
- After warm-up and learning-rate reduction, the loss settles into a low range;
- Validation loss follows the training loss with only a moderate difference.

This pattern suggests stable learning of the relationship between phonemes, spectrograms, and prosodic targets. Warm-up and learning-rate reduction are suitable because the model learns several related objectives simultaneously: phoneme duration, F0, loudness, and mel-spectrogram reconstruction.

4.2 Training Instability and Overfitting

There is no clear evidence of serious overfitting, although a few observations should be considered:

- Telugu validation loss varies more noticeably than English validation loss, reflecting its more challenging phonology and prosody;
- A training-validation difference exists, but it is not large, indicating generalization without extensive memorization;
- Minor fluctuations near the end of training may occur as the model balances prosody estimation with spectral reconstruction.

This behaviour is typical of multi-objective TTS models, which must balance audio quality, prosody, and alignment precision. If training is extended, additional regularization or further fine-tuning may help reduce the remaining variation.

4.3 English and Telugu Convergence

English converges more consistently because its task and acoustic patterns are more uniform. Telugu is more difficult due to wider phonetic diversity, script-dependent prosodic patterns, more complex vowels and syllables, and greater reliance on language-specific timing and intonation. As a result, Telugu generally converges more slowly and may show greater waveform-level variation on challenging sentences.

5. Evaluation of Synthesized Speech

Generated speech for five test utterances in each language is saved in Results/English/test_generated_audios/ and Results/Telugu/test_generated_audios/.

5.1 English Test Utterances

#	Utterance	Assessment
1	Hello, how are you today?	Pronunciation is clear and natural, with good pacing. The final intonation is slightly flat, but the speech remains smooth and understandable.
2	The researchers will conduct an experiment.	The sentence is fully understandable and follows the phrase structure. The word “experiment” has slight roughness, with stress and timing occasionally compressed.
3	Students learn new things every day.	Very clear speech with stable prosodic flow and balanced phrase separation; this is among the most natural English samples.
4	And they went off into the space when earth was burning up or being destroyed	This is the most difficult English item. Its long and complex phrasing causes prosodic compression, rushed words, and unstable intonation. The meaning remains clear, but naturalness is reduced.
5	Karnataka's expansive list of handmade artisanal products are impressive	The sentence is mostly understandable, but its rhythm resembles hesitation and its emphasis is unnatural. The long, vocabulary-dense structure reduces smoothness and exposes limitations with complex sentences.

5.2 Telugu Test Utterances

Note: The Telugu sentences could not be recovered as readable script from the source document. They are therefore identified by number below; the original Telugu text can be inserted in the indicated places.

#	Utterance	Assessment
1	[Telugu greeting phrase – insert original]	The greeting is clear and understandable, with natural opening and closing tones. Pitch variation is strong but remains acceptable.
2	[Telugu sentence 2 – insert original]	Very clear and natural output. This short sentence has stable timing and strong intelligibility and is one of the best Telugu results.
3	[Telugu sentence 3 – insert original]	Overall quality is good, with mostly natural phrase rhythm. Some emphasis is slightly overstated, but the delivery remains fluent.
4	[Telugu sentence 4 – insert original]	The output is generally understandable and expressive, and timing follows the clause structure. Phrasing is slightly robotic at sentence boundaries, but the content remains clear.
5	[Telugu sentence 5 – insert original]	Speech is natural and smooth, with good clarity and phrase flow. It is more stable and easier to follow than the long English sentence.

5.3 Overall Speech Quality

For both languages, short sentences with simple syntax produce the most natural speech. Quality decreases most for long, complicated, or heavily inflected sentences. Sentence length affects both languages, while Telugu relies more strongly on stable phrase-level delivery because of its prosodic structure and spelling complexity.

6. Prosody-Predictor Plots: Timing, F0, and Loudness

Visualizations for each evaluation utterance are stored in Results//variance_adaptor_plots/ under the duration/, pitch/, and energy/ subdirectories. These plots illustrate how the model allocates duration, F0, and loudness to phonemes or syllables.

6.1 Phoneme Duration

In both languages, content words and stressed syllables generally receive longer durations, while function words and short connecting words receive shorter durations. Words at the ends of sentences are commonly lengthened. In English utterance 1, duration increases around “hello”, “today”, and the final question phrase. In utterance 2, “conduct an experiment” lasts longer than nearby function words, reflecting natural emphasis. Telugu duration varies more because its prosody is strongly affected by word form and accent placement. The greeting (utterance 1) and utterance 4 show longer durations for content-rich words, while function-like segments remain short.

6.2 Fundamental Frequency (Pitch)

F0 is estimated for each phoneme and normalized. Common patterns include higher F0 at the beginning of words or on emphasized syllables, a gradual decline across a phrase unless a new idea or question requires a rise, and a more noticeable pitch contour near the end of questions. English utterance 1 rises in its opening phrase and falls toward the end, as expected in a conversational question. Utterance 2 is flatter and has a neutral declarative delivery. Telugu plots show stronger local variation: the greeting has a distinct oscillating contour typical of a question, while utterance 5 progresses more evenly, matching a mild declarative rhythm.

6.3 Loudness (Energy)

Loudness acts as an indicator of emphasis, with higher values around stressed words, clause boundaries, and important vocabulary.

- English utterance 2 has loudness peaks around “conduct” and “experiment”, representing reasonable semantic emphasis;
- English utterance 4 displays uneven loudness because its long and dense sentence is difficult to interpret;
- Telugu utterances 3 and 5 have strong peaks on content words, supporting natural emphasis, although local intensity is occasionally too high.

6.4 Interpretation of the Plots

The visualizations indicate that the model learns useful prosodic patterns: longer durations for content words, F0 changes around phrase boundaries, and stronger loudness on meaningful words. They also expose limitations. Long sentences produce compressed or unstable contours, complex syntax breaks up the loudness and F0 patterns, and Telugu prosody remains more variable and less stable than English because of script-specific phonology and vocabulary complexity.

7. English and Telugu: Comparative Analysis

7.1 English Advantages

English achieves better overall results. Its dataset and preprocessing are cleaner and more consistent, sentence-level prosody is more uniform with fewer script-related phonological ambiguities, and the reference implementation is naturally suited to the language. Consequently, English samples sound smoother, more natural, and more rhythmically consistent.

7.2 Telugu Advantages

The Telugu model performs well on short sentences with a clear structure. It captures stress and speech rhythm reasonably well when phrases are compact, and short greetings or simple statements can sound clear and natural.

7.3 Constraints and Failure Cases

- Long sentences or sentences with complex syntax can produce unnatural rhythm;
- Pronunciation quality may decline for uncommon vocabulary patterns or dense phrasing;
- F0 or loudness values may be unstable for some words;
- Telugu is more affected by morphological and prosodic complexity;
- Speech may sound robotic or discontinuous when duration and F0 are not properly aligned. These limitations are common in small-scale TTS systems.

7.4 Recommended Enhancements

- Expand the Telugu dataset with more varied speech samples;
- Increase training duration or fine-tune using prosody-focused loss terms;
- Use more reliable phoneme lexicons and Telugu text normalization;
- Experiment with alternative vocoder configurations and acoustic-model variants;
- Improve tokenization and normalization for punctuation and script-specific behaviour;
- Test on more than five representative sentences;
- Include objective measures such as MOS, intelligibility, and F0 error along with listening-based evaluation.

8. Reproducibility

8.1 Required Libraries

The required packages are specified in project/requirements.txt, with separate language-level files at project/English/requirements.txt and project/Telugu/requirements.txt. The software environment follows the usual FastSpeech 2 stack, including PyTorch and audio-processing and machine-learning libraries.

8.2 Configuration Files

The directories project/English/config/English/ and project/Telugu/config/Telugu/ contain preprocess.yaml, model.yaml, and train.yaml. These files specify dataset locations, feature extraction, F0 and loudness normalization, training parameters, and model architecture.

8.3 Execution Commands

Installation

```
cd project
pip install -r requirements.txt
```

English

```
cd project/English
python prepare_align.py config/English/preprocess.yaml
python preprocess.py config/English/preprocess.yaml
python train.py -p config/English/preprocess.yaml -m config/English/model.yaml -t config/English/train.yaml
python synthesize.py --text "Hello, how are you today?" --restore_step 40000 --mode single -p
config/English/preprocess.yaml -m config/English/model.yaml -t config/English/train.yaml
```

Telugu

```
cd project/Telugu
python prepare_align.py config/Telugu/preprocess.yaml
python preprocess.py config/Telugu/preprocess.yaml
python train.py -p config/Telugu/preprocess.yaml -m config/Telugu/model.yaml -t config/Telugu/train.yaml
```

```
python synthesize.py --text "" --restore_step 40000 --mode single -p config/Telugu/preprocess.yaml -m
config/Telugu/model.yaml -t config/Telugu/train.yaml
```

These instructions follow the project README and the configuration structure maintained separately for each language.

9. Conclusion

This project demonstrates that one shared FastSpeech 2 architecture can support both English and Telugu when paired with language-specific preprocessing and configuration. English speech is more consistent and natural, while Telugu shows encouraging results but is more affected by prosodic complexity and long sentences.

The prosody-predictor visualizations indicate that the model learns meaningful duration, F0, and loudness patterns. However, the difference in quality between short and long sentences suggests that additional training data, longer training, and improved language-specific normalization could enhance intelligibility and naturalness in both languages. The repository is organized clearly, and its configuration files make training and inference settings accessible. With more diverse data and an extended training schedule, both models have scope for substantial improvement.